Online Retail II — Exploratory Data Analysis

ITSimplera Institute Internship Task Documentation Report

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Dataset: Online Retail II (UCI ML Repository)

Tools: Python, Pandas, Matplotlib, Seaborn — VS Code (Jupyter Notebook)

1. Objective

The goal of this task was to explore the Online Retail II dataset — real transaction data from a UK-based online retailer covering 2009 to 2011 — and produce a complete Exploratory Data Analysis (EDA) before the company invests in building a recommendation system. The analysis needed to answer three core business questions: who the customers are, which products sell the most, and which countries generate the most revenue. This stage was strictly observation-only — no rows were removed or modified at any point; missing values, duplicates, and outliers were identified and visualized, but the dataset was kept fully intact.

2. About the Dataset

The dataset was sourced from the UCI Machine Learning Repository. It contains invoice-level transaction records with the following columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, CustomerID, and Country. The original file ships as two Excel sheets ('Year 2009-2010' and 'Year 2010-2011'), which were loaded and combined into a single DataFrame before analysis.

3. Environment & Setup

The notebook was built and executed in VS Code using the Jupyter extension, with Python 3.11. Core libraries used were pandas and numpy for data handling, and matplotlib and seaborn for visualization.

4. Loading the Dataset & Basic Information

Both sheets of the Excel file were read and concatenated into one DataFrame. After loading, the dataset was inspected for its shape, column names, and data types before any further analysis.

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Dataset loaded successfully.
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	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	CustomerID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13,085.00	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13,085.00	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13,085.00	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13,085.00	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13,085.00	United Kingdom

Figure 1: First 5 rows of the combined dataset after loading.

The combined dataset came out to:

- 1,067,371 rows and 8 columns
- Columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, CustomerID, Country

	count	unique	top	freq	mean	min	25%
Invoice	1067371	53628	537434	1350	NaN	NaN	NaN
StockCode	1067371	5305	85123A	5829	NaN	NaN	NaN
Description	1062989	5698	WHITE HANGING HEART T-LIGHT HOLDER	5918	NaN	NaN	NaN
Quantity	1,067,371.00	NaN	NaN	NaN	9.94	-80,995.00	1.00
InvoiceDate	1067371	NaN	NaN	NaN	2011-01-02 21:13:55.394029	2009-12-01 07:45:00	2010-07-09 09:46:00
Price	1,067,371.00	NaN	NaN	NaN	4.65	-53,594.36	1.25
CustomerID	824,364.00	NaN	NaN	NaN	15,324.64	12,346.00	13,975.00
Country	1067371	43	United Kingdom	981330	NaN	NaN	NaN

Figure 2: Summary statistics (df.describe()) across all columns.

5. Missing Values

Missing values were checked per column, without dropping anything. Two columns had missing data:

Missing Values Report:

	Missing Count	Missing %
CustomerID	243007	22.77
Description	4382	0.41

Figure 3: Percentage of missing values by column.

CustomerID missing in almost a quarter of all transactions is the more significant gap here — since a recommendation system needs to know who bought what, these 243,007 unlinked transactions currently can't be used for personalization at all.

6. Duplicate Rows

Exact duplicate rows were checked using df.duplicated(). A total of 34,335 rows (3.22% of the dataset) were found to be exact duplicates across all columns. These were not removed, in line with the observation-only requirement, but they are flagged here as something to clean before any future modeling step, since duplicate rows could inflate purchase counts for certain products.

7. Top 10 Best-Selling Products by Quantity

Products were grouped by Description and summed by total Quantity sold to find the top 10 best-sellers by volume.

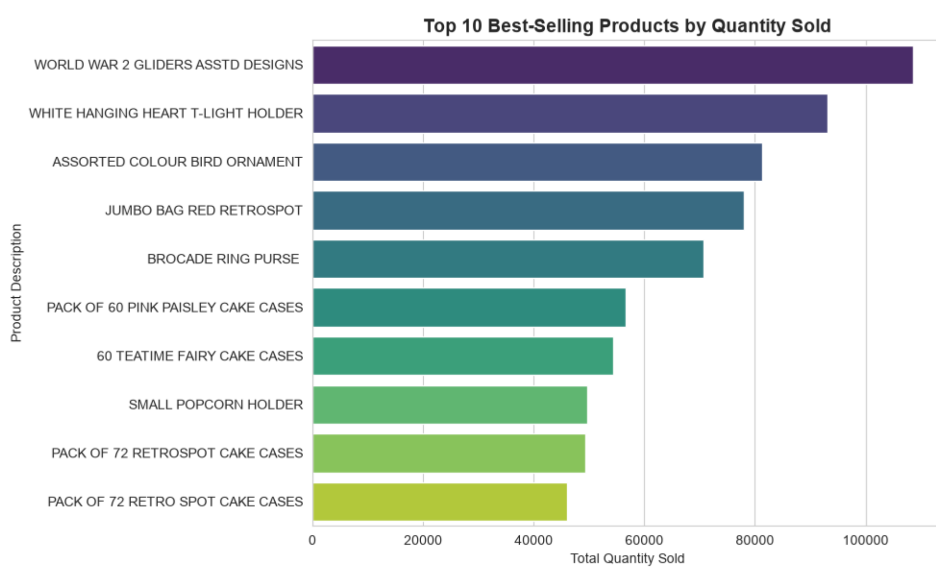

Figure 4: Top 10 products by total quantity sold.

"WORLD WAR 2 GLIDERS ASSTD DESIGNS" was the single most-purchased product by volume, with 108,545 units sold, followed by "WHITE HANGING HEART T-LIGHT HOLDER" at 93,050 units.

8. Top 10 Best-Selling Products by Revenue

The same grouping was repeated using a calculated Revenue column (Quantity × Price) instead of raw quantity, to see which products actually earn the most money.

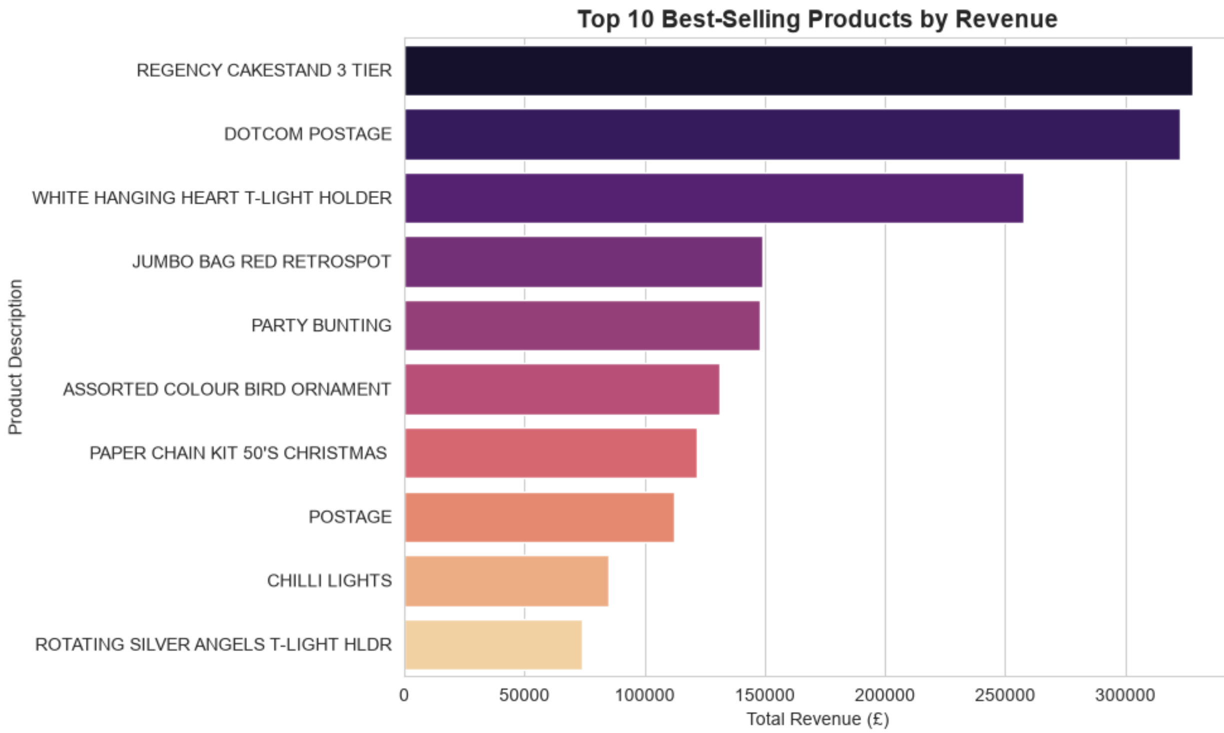

Figure 5: Top 10 products by total revenue.

"REGENCY CAKESTAND 3 TIER" generated the most revenue at £327,813.65, closely followed by "DOTCOM POSTAGE" at £322,647.47. Interestingly, the top product by quantity (WORLD WAR 2 GLIDERS) doesn't even appear in this top 10 by revenue — showing that high-volume products aren't necessarily the highest earners.

9. Sales Performance by Country

Transactions were grouped by Country to compare total revenue, order count, and quantity sold across markets.

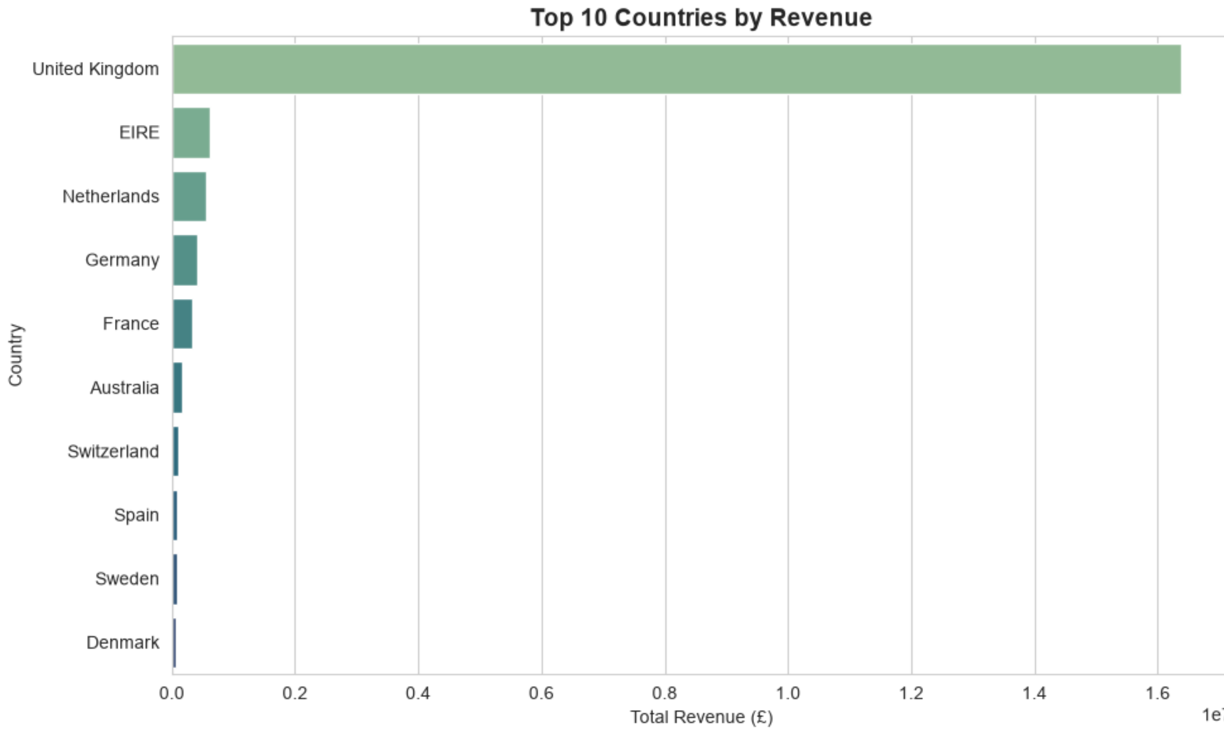

Figure 6: Top 10 countries by total revenue.

The United Kingdom dominates, generating £16,382,583.90 in revenue — approximately 85% of the business's total revenue — from 49,108 unique orders. The next closest market, EIRE, brought in just £615,519.55. This shows the business is heavily concentrated in a single domestic market, which is worth flagging as a geographic concentration risk.

10. Revenue Over Time — Monthly Trend

InvoiceDate was converted to a proper datetime type and grouped by month to plot the revenue trend across the full 2009–2011 period.

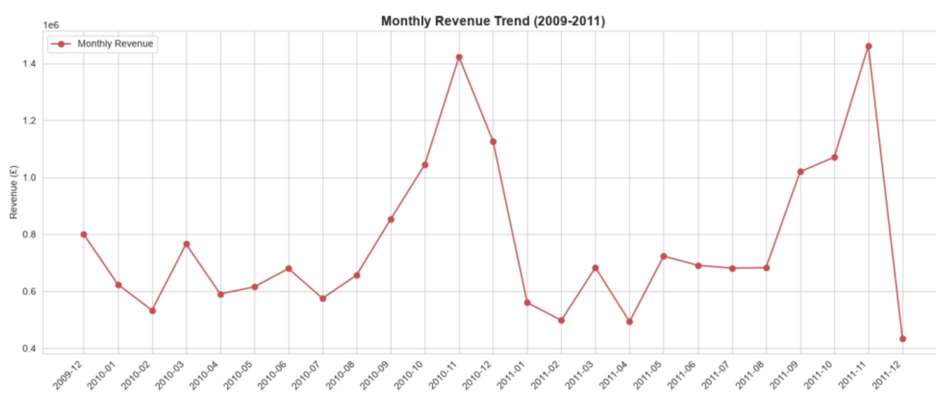

Figure 7: Monthly revenue trend from December 2009 to December 2011.

Revenue shows a clear seasonal pattern, climbing steadily from mid-year and peaking sharply in November both years — £1.42M in November 2010 and £1.46M in November 2011 — before dropping off in December. This lines up with pre-Christmas holiday shopping. The sharp drop shown for December 2011 is most likely because the dataset ends partway through that month rather than a genuine dip in demand.

11. Correlation Heatmap

A correlation heatmap was built across the numeric columns: Quantity, Price, Revenue, and CustomerID.

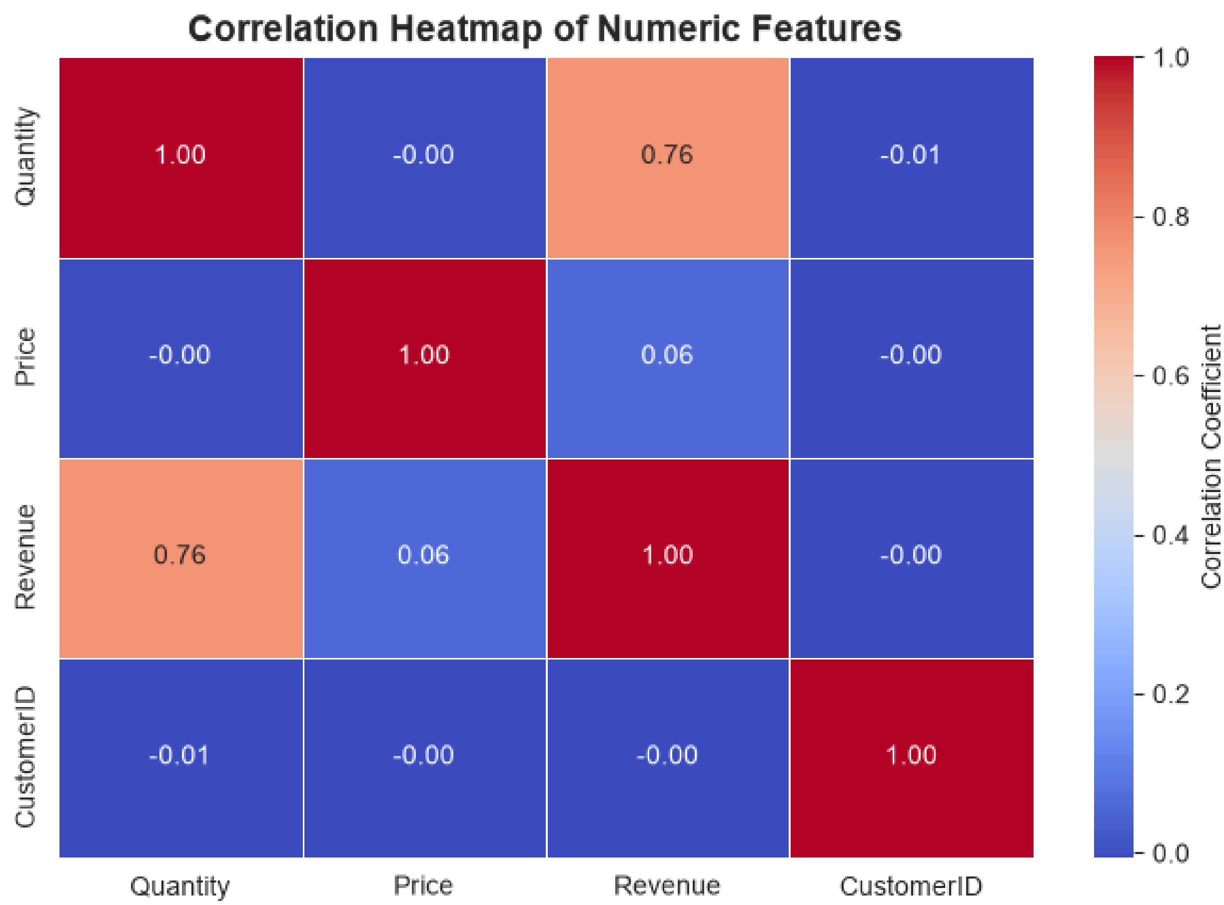

Figure 8: Correlation heatmap of numeric features.

Quantity and Revenue are strongly correlated (0.76), which makes sense since Revenue is directly derived from Quantity. Price and Revenue, on the other hand, show almost no correlation (0.06) — meaning revenue in this business is driven far more by how much customers buy rather than by how expensive individual items are. CustomerID shows no meaningful correlation with anything, which is expected since it's an identifier, not a real numeric feature.

12. Outlier Detection — Box Plots

Box plots were used to visually inspect Quantity and Price for outliers, and the IQR method was used to quantify how many outliers exist in each column.

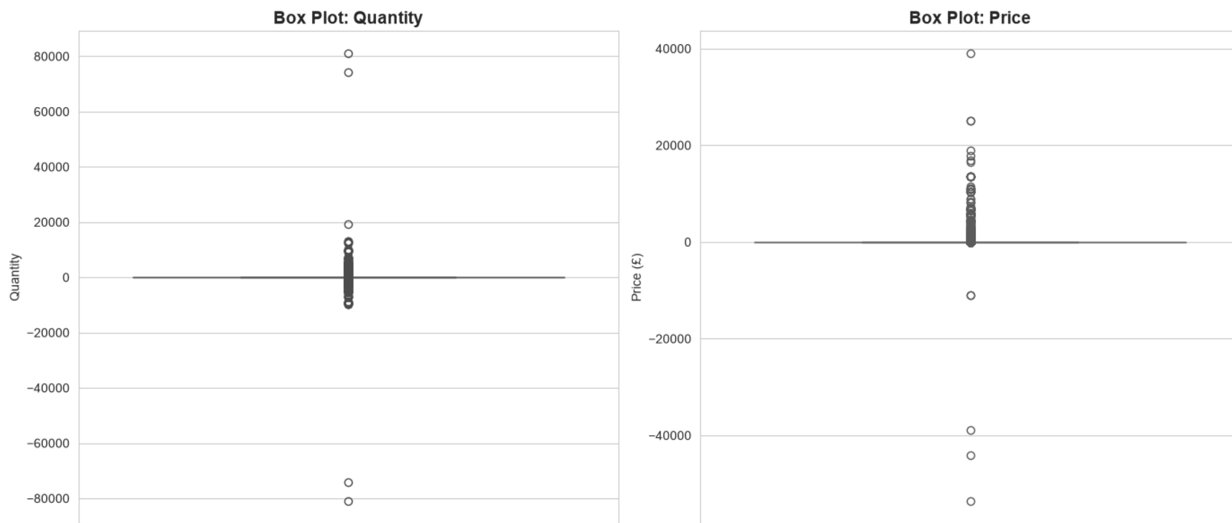

Figure 9: Box plots for Quantity and Price showing extreme values on both ends.

Both columns show extreme values in both directions — Quantity ranges from -80,995 to +80,995, and Price ranges from -£53,594.36 to +£38,970. These aren't random data entry errors; they reflect genuine bulk/wholesale orders and their corresponding cancellations, since negative quantities in this dataset represent cancelled or returned orders (Invoice numbers starting with 'C'). This is an important point for future modeling — a standard IQR-based outlier removal would strip out legitimate high-value transactions along with actual noise, so any cleaning step needs to treat cancellations and bulk orders differently from genuine errors.

13. Business Insights

Based on the full analysis above, here are the key takeaways for the business:

1. Revenue and volume leaders aren't the same product

"WORLD WAR 2 GLIDERS ASSTD DESIGNS" sold the most units (108,545), but "REGENCY CAKESTAND 3 TIER" generated the highest revenue (£327,813.65) despite not appearing in the top 10 by quantity at all. A recommendation engine optimized purely for "most purchased" would completely miss the products that actually drive the most money.

2. The business is almost entirely UK-dependent

United Kingdom alone contributed roughly 85% of total revenue (£16,382,583.90) across all 43 countries in the dataset. The next closest market, EIRE, brought in only £615,519.55. This heavy concentration is a risk — any slowdown in the UK market would hit the business far harder than a geographically diversified retailer, and it points to real growth potential in underdeveloped markets like Germany, France, and the Netherlands.

3. Revenue is strongly seasonal, peaking around November

Monthly revenue climbs steadily from mid-year and peaks sharply in November both years (£1.42M in 2010, £1.46M in 2011) before dropping — consistent with pre-Christmas gift buying. Inventory planning, staffing, and marketing budget should be weighted toward Q4, especially October–November, rather than spread evenly across the year.

4. Nearly a quarter of transactions can't be tied to a customer

CustomerID is missing in 22.77% of all rows (243,007 transactions). Since a recommendation system depends entirely on linking purchases to a specific customer, this is a significant blind spot — almost 1 in 4 transactions currently can't feed into personalized recommendations at all, and this gap needs to be understood (guest checkouts? data entry issue?) before modeling begins.

5. Outliers are business behavior, not bad data

Quantity ranges from -80,995 to +80,995, and Price ranges from -£53,594.36 to +£38,970 — extreme values on both ends. These reflect large wholesale/bulk orders and their corresponding cancellations rather than random errors. Naively removing "outliers" by a standard IQR cutoff would strip out legitimate high-value B2B transactions along with genuine noise.

6. Quantity, not price, drives revenue

The correlation heatmap shows Quantity and Revenue are strongly correlated (0.76), while Price and Revenue are barely correlated (0.06). Revenue in this business is driven much more by how much customers buy than by how expensive individual items are — suggesting volume-based promotions (bundles, multi-buy discounts) may be more effective than price-based ones.

7. Duplicate records exist but are limited

34,335 rows (3.22% of the dataset) are exact duplicates. This is moderate but non-trivial — while the task required keeping them for this observation-only stage, they should be flagged for cleaning before any modeling step, since duplicate transactions could artificially inflate perceived purchase frequency for certain products or customers.

14. Conclusion

This EDA gave a clear, data-backed picture of the retailer's business before any modeling work begins. The company is heavily UK-centric, strongly seasonal around the holiday period, and its top-selling products by volume are not always its top earners. The most important finding for the planned recommendation system is the CustomerID gap — nearly a quarter of transactions can't currently be linked to a customer, which will need to be addressed before personalized recommendations can be built reliably. No data was removed at this stage, keeping the analysis true to the original dataset as required.